Agentic Geo-VQA

A Multimodal Vision-Language Architecture for Spatiotemporal Landslide Susceptibility Forecasting

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Presentation Outline

1. Introduction & Motivation
2. Study Area & Dataset Analysis
3. Methodology: Agentic Geo-VQA Architecture

Motivation & Geohazard Context

- **Global Impact:** Landslides are devastating natural hazards causing profound socio-economic and infrastructural losses.
- **Regional Focus:** Tropical, monsoon-dominated environments such as the Raigad District, Maharashtra.
- **Aggravating Mechanisms:**
  - Steep structural topography inherent to the Western Ghats.
  - Aggressive anthropogenic deforestation and land-use shifts.
  - Intense, prolonged seasonal monsoon precipitation.
- **The Core Need:** Accurate Landslide Susceptibility Mapping (LSM) is a critical prerequisite for effective spatial planning and disaster risk reduction.

Limitations of Traditional Models

- **Historical Evolution:** Predictive mapping has advanced from basic statistical models to sophisticated Deep Learning architectures (CNNs, standard LSTMs).
- **The "Black Box" Problem:**
  - Conventional models output simple binary classifications or isolated probabilities (e.g., $P(\text{failure}) = 0.85$).
  - They fail to articulate the synergistic interactions of geomorphological and temporal factors driving the prediction.
- **The Core Limitation:** They predict *where* a slope might fail, but cannot explain *why*.
- **Our Objective / The Paradigm Shift:** Transitioning from basic probabilistic mapping into Agentic Geospatial Visual Question Answering (Geo-VQA) to provide interpretable, diagnostic geotechnical intelligence.

Study Area: Raigad District, Maharashtra

Geographic Setting

- Located along the Western Ghats escarpment, western India.
- Coastal monsoon district: mean annual rainfall exceeding 3000 mm.
- Historically one of the **highest landslide-fatality** districts in Maharashtra.

Dataset Composition

- **19 static geospatial rasters:** Elevation, Slope, Aspect, TWI, SPI, Curvature, Soil Groups, Distance to Lineaments, and more.
- **8-year temporal LULC** sequence: 2017–2024 (satellite-derived).
- **Multi-station rainfall** records for meteorological triggering analysis.

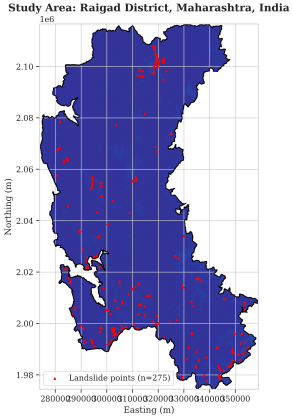

Figure: Raigad District: Landslide inventory and geospatial domain.

Topographic & Geological Conditioning Factors

19 Static Geospatial Rasters Extracted:

- Elevation, Slope, Aspect
- Profile & Plan Curvature
- Topographic Wetness Index (TWI)
- Stream Power Index (SPI)
- Distance to Lineaments
- Distinct Soil Groups

Key Structural Disparities:

Metric	Background	Landslide
Elev. Mean (m)	113.9	242.3
Slope Mean (°)	9.2	24.1
Slope Max (°)	50.97	59.29

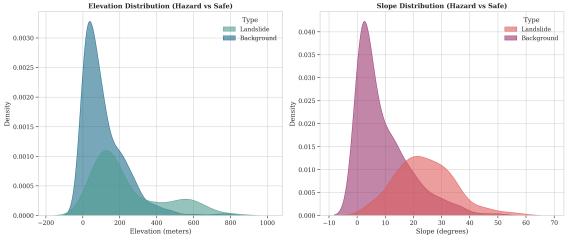

Figure: Probability density distributions of key topographic factors across Raigad District.

Temporal LULC Dynamics (2017–2024)

Why LULC Matters:

- Vegetation root systems provide biological anchoring and control soil cohesion.
- Progressive deforestation removes this natural stabilizer, increasing susceptibility to mass wasting during monsoon saturation.

Critical 8-Year Shifts:

Class	2017 (%)	2024 (%)
Rangeland / Shrubland	39.77	47.13
Dense Forest / Trees	37.02	29.58
Agricultural Crops	14.86	12.01
Built-up / Urban	4.70	8.10

Net result: -7.4% dense forest cover in 7 years.

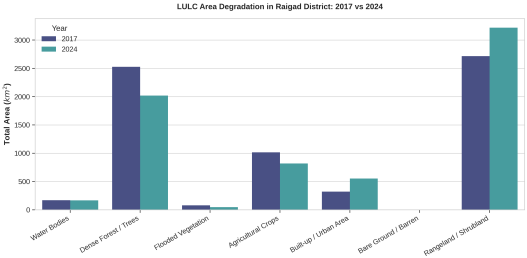

Figure: LULC area degradation in Raigad District: 2017 vs 2024.

Meteorological Triggers: Rainfall Analysis

Rainfall as the Primary Trigger:

- Heavy precipitation increases pore water pressure within the soil matrix.
- Rising pore pressure reduces effective normal stress, directly violating the Mohr-Coulomb failure criterion:

$$\tau_f = c' + (\sigma - u)\tan\phi'$$

- Above a critical rainfall threshold, even slopes with modest gradients become unstable.

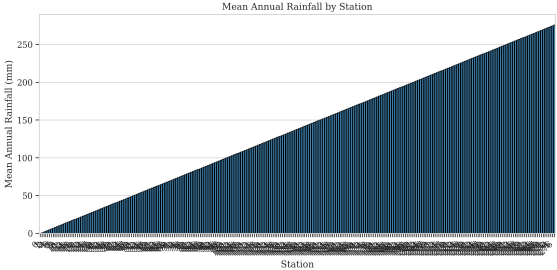

Figure: Mean decadal rainfall across Raigad stations showing intense coastal precipitation zones.

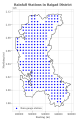

Figure: Spatial distribution of meteorological stations.

Methodology Overview: The Agentic Geo-VQA Pipeline

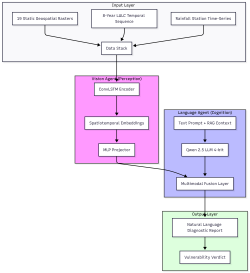

Figure: End-to-end Agentic Geo-VQA architecture fusing a spatiotemporal vision encoder with a cognitive language agent.

Agent 1
Spatiotemporal Vision Agent
ConvLSTM encoder processes 8-year geospatial patch sequences.

Bridge
MLP GeoProjector
Projects 64-dim visual embeddings into the 3584-dim LLM word space.

Agent 2
Cognitive Language Agent
Qwen 2.5 7B LLM generates expert geotechnical diagnostic reports.

Spatiotemporal Vision Encoding: ConvLSTM

Why ConvLSTM?

- Standard LSTMs flatten spatial inputs, destroying local terrain context.
- ConvLSTM replaces all matrix multiplications with **convolution operators** ($*$), preserving the 32×32 spatial grid at every time step.
- 4 stacked layers with hidden dims `[64, 128, 128, 64]`.

Gate Equations at time step t:

$$i_t = \sigma(W_{xi} * \mathcal{X}_t + W_{hi} * \mathcal{H}_{t-1} + W_{ci} \circ \mathcal{C}_{t-1} + b_i)$$

$$f_t = \sigma(W_{xf} * \mathcal{X}_t + W_{hf} * \mathcal{H}_{t-1} + W_{cf} \circ \mathcal{C}_{t-1} + b_f)$$

$$\mathcal{C}_t = f_t \circ \mathcal{C}_{t-1} + i_t \circ \tanh(W_{xc} * \mathcal{X}_t + W_{hc} * \mathcal{H}_{t-1} + b_c)$$

$$o_t = \sigma(W_{xo} * \mathcal{X}_t + W_{ho} * \mathcal{H}_{t-1} + W_{co} \circ \mathcal{C}_t + b_o)$$

$$\mathcal{H}_t = o_t \circ \tanh(\mathcal{C}_t)$$

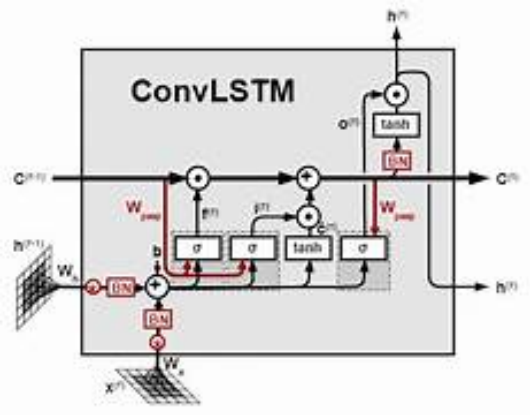

Figure: Internal cell architecture of the ConvLSTM network.

Cognitive Language Agent: Qwen 2.5 7B + QLoRA

The Language Agent (Qwen 2.5 7B-Instruct):

- Translates high-dimensional visual embeddings into expert geotechnical diagnostic text.
- Governed by scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Hardware Constraint: 16 GB NVIDIA Tesla P100 GPU.

Solution — QLoRA:

- Freezes pre-trained weights $W \in \mathbb{R}^{d \times k}$.
- Injects low-rank adapters $A \in \mathbb{R}^{r \times k}$, $B \in \mathbb{R}^{d \times r}$, $r \ll \min(d, k)$.
- Modified forward pass:

$$h = Wx + \frac{\alpha}{r} BAx$$

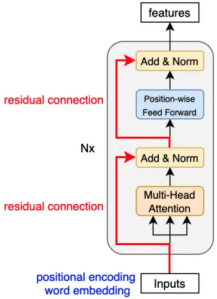

Figure: Transformer decoder architecture of the Qwen 2.5 language model.

Quantitative Results: Model Performance

Spatiotemporal Vision Agent (ConvLSTM)

Evaluated on unseen held-out test set:

Metric	Score
Accuracy	87.95%
Precision	0.8780
Recall	0.8780
F1-Score	0.8780
AUC-ROC	**0.9535**
MCC	0.7590

Topographic Benchmark Disparities:

Class	Elev. Mean	Slope Mean	Slope Max
Background	113.9 m	9.2°	50.97°
Landslide	242.3 m	24.1°	59.29°

Multimodal Alignment Convergence:

Training Phase	CLM Loss
Epoch 1 — Batch 1	3.4380
Epoch 1 — Batch 162	∼2.10
Epoch 2 — Batch 324	1.4264

Key Takeaways:

- An AUC-ROC of 0.9535 places the ConvLSTM encoder among publishable state-of-the-art LSM results.
- The CLM loss convergence from 3.44 to 1.43 in only 2 epochs confirms that the visual and linguistic vector spaces successfully synchronised.
- Mixed-precision (AMP) + Gradient Checkpointing kept peak VRAM within the 16 GB P100 budget.

Qualitative Results: Agentic Inference Case Studies

Case 1: HIGH Susceptibility

Terrain: Elevation 242.3 m, Slope 24.1°

Dominant LULC: Rangeland / Shrubland (47.13%)

"The elevation of this patch is notably higher than the regional background average, suggesting increased gravitational potential energy. The local slope gradient is more than twice as steep, amplifying shear stress and reducing stability margins along the failure plane."

Verdict: **HIGH Vulnerability** — Steep topography combined with inadequate biological soil anchoring creates an imminent geohazard risk under saturation conditions.

Case 2: LOW Susceptibility

Terrain: Elevation 113.9 m, Slope 9.2°

Dominant LULC: Dense Forest / Trees (29.58%) + Agricultural Crops (12.01%)

"The slope gradient is mild, recorded at 9.23 degrees. At this inclination, gravitational shear stress on the soil profile remains well within safe stability thresholds. Mature, deep root systems provide excellent biological anchoring, enhancing soil cohesion and regulating hydrological runoff."

Verdict: **LOW Vulnerability** — Terrain maintains robust geomorphological stability.

The system explains *why* a terrain is susceptible — not just *whether* it is.

Conclusion & Academic Significance

Key Contributions:

- **Paradigm shift:** From black-box probabilistic LSM to Agentic Geo-VQA — interpretable diagnostic geotechnical intelligence.
- **ConvLSTM vision agent** with an AUC-ROC of 0.9535, capturing 8 years of spatiotemporal terrain dynamics at 32×32 patch resolution.
- **QLoRA-fine-tuned Qwen 2.5 7B** deployed on a 16 GB GPU via 4-bit NF4 quantisation and low-rank adaptation.
- **MLP GeoProjector** bridges a 64-dim visual embedding into a 3584-dim linguistic space, enabling fluent geotechnical report generation.
- **CLM loss** converged from 3.44 to 1.43 in only 2 epochs, confirming multimodal alignment.

Future Directions:

- Integration of real-time rainfall telemetry for dynamic triggering alerts.
- Extension to SAR and optical multi-sensor fusion for cloud-penetrating monitoring.
- Scaling the framework district-wide for state-level Early Warning Systems.

Thank You

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